

Anas Javaid

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EDUCATION

Western University

London, ON

Bachelor of Electrical Engineering (CO-OP)

Sep 2024 – May 2028

- Dean's Honour List (2025–2026).
- Relevant Courses: Digital Logic Systems, Electronic Circuits, Control Systems, Data Structures & Algorithms, Linear Algebra, Power Systems.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Matlab, Java, JavaScript, HTML/CSS

Skills: Altium Designer, Arduino, Git, AutoCAD, Soldering, Microsoft VS Code, Fusion 360, 3-D printing.

PROJECTS

Reaction Time Trainer | C++, Altium, Arduino

Feb 2026 - Apr 2026

- Led a 3-person team to develop an Arduino Uno-based reaction time system with 3 interactive game modes.
- Delivered multi-modal gameplay by integrating **4** sensors and **5** actuators including a joystick, I2C LCD, buzzer, and servo, resulting in responsive, real-time user feedback.
- Achieved a system response latency of **under 50 ms** by developing C++ firmware for real-time input polling, randomized signal generation, and instant performance feedback.
- Designed and soldered a single-layer **PCB shield** in **Altium**, optimizing layout and trace routing for reliability.

Emergency Route Planner | HTML/CSS, JavaScript

Jan 2025 - Apr 2025

- Developed a web-based emergency route planning tool using **Leaflet.js** and **Turf.js** to calculate safe navigation paths during natural disasters like floods and wildfires.
- Implemented routing logic using graph algorithms and GIS data to compute safest and fastest emergency routes.
- Integrated the **OpenRouteService API** to fetch road network data and deliver optimized alternative routes.

RC Car | C++, Arduino

Jul 2025 - Oct 2025

- Developed a Bluetooth-controlled RC car with **<178 ms** control latency using Arduino Nano, HC-05 module, H-bridge motor driver, and SG90 servo.
- Programmed motor control logic in C++ for forward, reverse, and turning maneuvers.
- Powered system with dual LiPo batteries and optimized **5V** distribution using buck converters.

Safe Security System | C++, Arduino

Aug 2025 - Oct 2025

- Built an Arduino Uno-based smart home security system integrating PIR, ultrasonic, and alarm actuators to deliver real-time alert triggers with zero failures across **15** validation runs.
- Implemented a 4-digit PIN authentication using an LCD status screen and SG90 servo locking mechanism, triggering a security alarm after 3 failed login attempts.
- Optimized system power efficiency by placing all peripherals in a low-power sleep state until triggered by PIR motion hardware interrupts, reducing power consumption by **77%**.

EXTRACURRICULAR ACTIVITIES

WeBots(Western Robotics Club)

Sep 2025 - Present

- Implemented real-time sensor integration across **SPI** and **I2C** buses using ESP32, capturing high rate feedback from IMUs and rotary encoders for closed loop motion control.
- Designed hardware **PID feedback loops** using operational amplifiers and H-bridge driver stages to regulate DC motor speed and torque, improving response time by **40%**.

Western Aero Design

Sep 2025 - Present

- Designed multi-layer PCBs in Altium, optimizing trace routing and ground planes for flight integrity.
- Collaborated with mechanical and software teams to ensure system functionality and safety compliance.